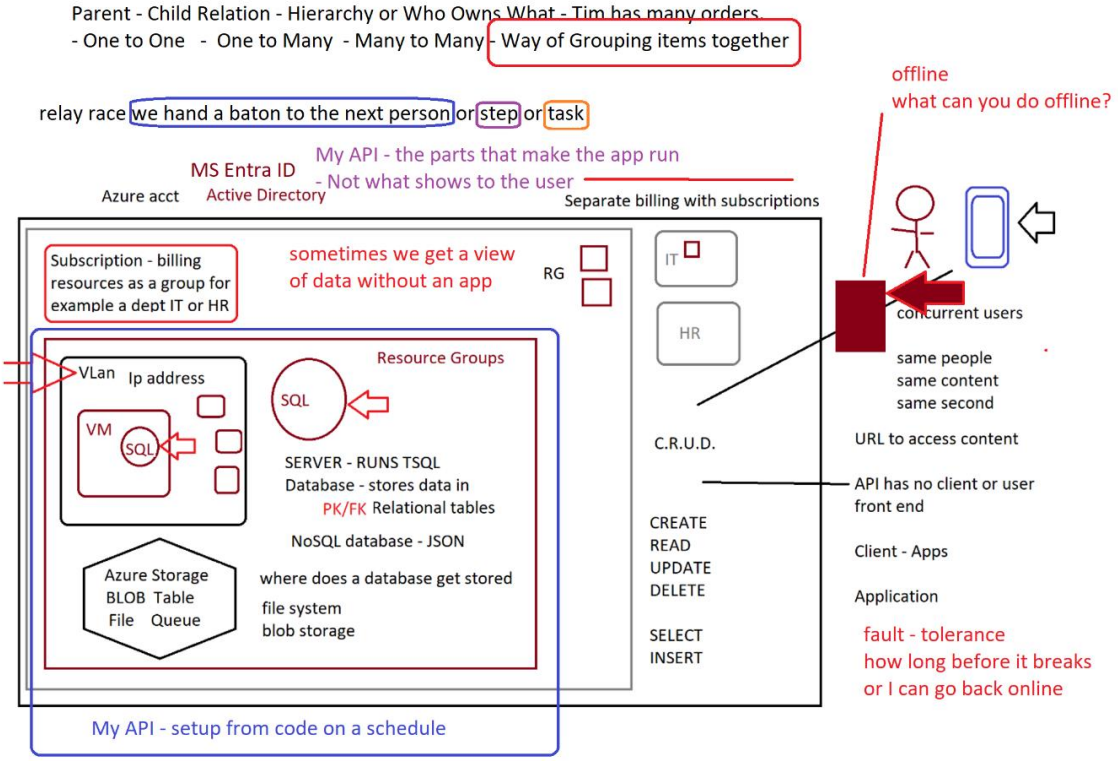
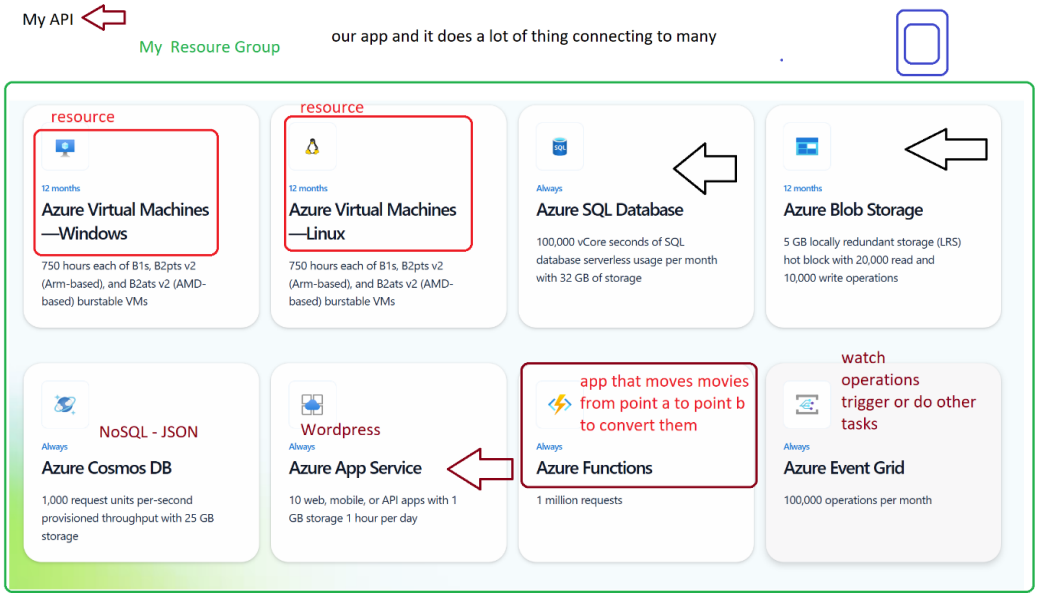
PBI Extras 202X





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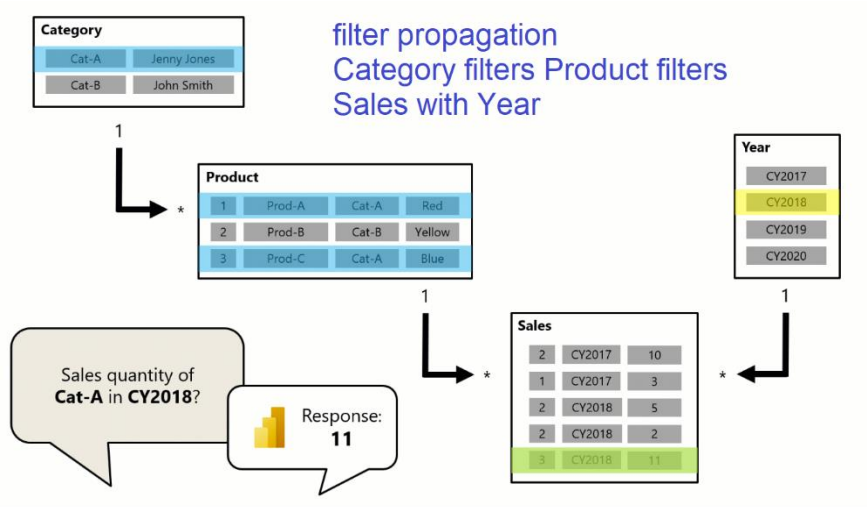


DAX Copy

```
CALCULATE(<expression>[, <filter1> [, <filter2> [, …]]])
```

Term	Definition
`expression`	The expression to be evaluated.
`filter1, filter2,…`	(Optional) Boolean expressions or table expressions that defines filters, or filter modifier functions.

The expression used as the first parameter is essentially the same as a measure.

Filters can be:

- Boolean filter expressions
- Table filter expressions
- Filter modification functions

Filter modifier functions

Filter modifier functions allow you to do more than simply add filters. They provide you with additional control when modifying filter context.

Function	Purpose
REMOVEFILTERS	Remove all filters, or filters from one or more columns of a table, or from all columns of a single table.
ALL [1], ALLEXCEPT, ALLNOBLANKROW	Remove filters from one or more columns, or from all columns of a single table.
KEEPFILTERS	Add filter without removing existing filters on the same columns.
USERELATIONSHIP	Engage an inactive relationship between related columns, in which case the active relationship will automatically become inactive.
CROSSFILTER	Modify filter direction (from both to single, or from single to both) or disable a relationship.

DAX Copy

```
Blue Revenue =
CALCULATE(
    SUM(Sales[Sales Amount]),
    'Product'[Color] = "Blue"
)
```

Category	Sales Amount	Blue Revenue
Accessories	$1,272,057.89	$165,406.62
Bikes	$94,620,526.21	$8,374,313.88
Clothing	$2,117,613.45	$259,488.37
Components	$11,799,076.66	$803,642.10
Total	**$109,809,274.20**	**$9,602,850.97**

The CALCULATE function evaluates the sum of the **Sales** table **Sales Amount** column in a modified filter context.

A new filter is added to the **Product** table **Color** column—or, the filter overwrites any filter that's already applied to the column.

Adventureworks Example

In Power BI with AdventureWorks data, USERELATIONSHIP activates an inactive relationship (e.g., Shipping Date) to calculate metrics (e.g., Shipped Amount) alongside an active one (Order Date). Use a DAX measure with CALCULATE to temporarily override the model's active relationship, allowing sales reporting based on alternate dates.

Scenario: Sales by Order Date vs. Ship Date

In the AdventureWorks model, the FactInternetSales table often links to the DimDate table via OrderDateKey (Active) and ShipDateKey (Inactive).

1. **Active Relationship:** DimDate[DateKey]

FactInternetSales[OrderDateKey]

2. **Inactive Relationship:** DimDate[DateKey]

FactInternetSales[ShipDateKey]

DAX Measure Example

To calculate the total sales amount based on the **Shipping Date** rather than the order date, create this measure:

dax

```
Sales Amount by Ship Date =
CALCULATE(
    SUM(FactInternetSales[SalesAmount]),
    USERELATIONSHIP(FactInternetSales[ShipDateKey], 'DimDate'[DateKey])
)
```

Key Details

- **CALCULATE:** The USERELATIONSHIP function must be used within CALCULATE or CALCULATETABLE.
- **Override:** The measure forces the relationship, regardless of whether it is active or inactive in the model view.
- **Result:** This measure can be used in visuals to show sales by Shipping Date, while a standard SUM measure will still show Sales by Order Date.
- **Blank Values:** Note that if an order has not shipped, USERELATIONSHIP will return a BLANK value for the ship date.

Notable DAX

Using DAX more like T-SQL

PREVIOUS(), RUNNINGSUM() and MOVINGAVERAGE() functions to create comparison metrics between each fiscal year.

USERPRINCIPALNAME()

DAX function reference - DAX | Microsoft Learn

Filter Context

The CALCULATE function is a powerful function you can use to manipulate the filter context. The first argument takes an expression or a measure (a measure is just a named expression). Subsequent arguments allow modifying the filter context.

Dates & Fiscal Years

The CALENDARAUTO function returns a single-column table comprising date values. The "auto" behavior scans all data model date columns to determine the earliest and latest date values stored in the data model. It then creates one row for each date within this range, extending the range in either direction to ensure full years of data is stored.

This function can take a single optional argument that is the last month number of a year. When omitted, the value is 12, meaning that December is the last month of the year. In this case, 6 is entered, meaning that June is the last month of the year.

https://github.com/MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst/blob/Main/Instructions/Labs/04-create-dax-calculations.md

USERELATIONSHIP

USERELATIONSHIP function (DAX) - DAX | Microsoft Learn

Example: To calculate the sum of internet sales and allow slicing by ShippingDate instead of the traditional OrderDate, - **create measure** with USERELEATIONSHIP

Using USERELATIONSHIP in calculated columns requires CALCULATE to activate inactive relationships, but it is challenging due to context transition, which often triggers before the relationship change. To force it to work, you must use CALCULATE(..., USERELATIONSHIP(..), REMOVEFILTERS(..)) to ensure the correct filter context applies.

Key Considerations for Calculated Columns

- **Context Transition:** Calculated columns create a row context, and CALCULATE turns this into a filter context (context transition). This transition happens *before* USERELATIONSHIP takes effect, often resulting in incorrect data.

- **Required Formula Structure:** To properly use it, combine CALCULATE with USERELATIONSHIP and usually REMOVEFILTERS on the related table to override the default active relationship.
- **Alternative Solutions:** If USERELATIONSHIP proves too complex or fails, using {Link: LOOKUPVALUE https://community.powerbi.com/t5/Desktop/Calculated-columns-USERELATIONSHIP-or-LOOKUPVALUE/td-p/1197385} or DAX measures is often preferred.

Example Scenario (Date Relationship)
To calculate a delivery date year in a Sales table using an inactive relationship to a Date table:

dax

```
Delivery Year =
CALCULATE(
   SELECTEDVALUE('Date'[Year]),
   USERELATIONSHIP('Sales'[DeliveryDate], 'Date'[Date]),
   REMOVEFILTERS('Date')
)
```

REMOVEFILTERS is crucial to prevent the default "Order Date" relationship from conflicting with the "Delivery Date" activation

USERELATIONSHIP in DAX is super handy when you have **multiple relationships** between two tables but only one can be active at a time (Power BI enforces a single active relationship between any two tables). You use it inside CALCULATE to **temporarily activate an inactive relationship** for that calculation.

Below is a crisp, realistic example you can copy/paste.

Scenario

You've got a Sales fact table and a Calendar table:

- Sales[OrderDate] — when the order was placed
- Sales[ShipDate] — when the order shipped
- Calendar[Date] — your standard date dimension

Model design:

- Active relationship: Calendar[Date] → Sales[OrderDate] (Active ✅)
- Inactive relationship: Calendar[Date] → Sales[ShipDate] (Inactive ⛔)

Why? Because most of your reports are order-based, but sometimes you need to analyze by **ship date**.

Measures

1) Base measure

Plain Text

Total Sales :=

SUM (Sales[SalesAmount])

2) Orders by Order Date (active by default)

This just works because the active relationship is on OrderDate:

Plain Text

Sales (Order Date) :=

[Total Sales]

3) Orders by Ship Date (activate the inactive relationship)

Use USERELATIONSHIP to flip to Ship Date for this calculation:

Plain Text

Sales (Ship Date) :=

CALCULATE (

[Total Sales],

USERELATIONSHIP (Calendar[Date], Sales[ShipDate])

)

Put both measures in the same visual (e.g., a line chart with Calendar[Date] on the axis) to compare performance by order vs. ship dates.

Useful Variations

A) Dynamic toggle via disconnected slicer

Let users pick "Order Date" or "Ship Date" with a slicer.

1. Create a small disconnected table Date Mode:
    - Column: DateMode with values: "Order Date", "Ship Date"
2. Measure:

DAX

Sales (Toggle Date Context) :=

VAR mode =

SELECTEDVALUE ('Date Mode'[DateMode], "Order Date")

RETURN

SWITCH (

TRUE(),

mode = "Ship Date",

CALCULATE ([Total Sales], USERELATIONSHIP (Calendar[Date], Sales[ShipDate])),

-- default: Order Date

[Total Sales]

)

``

B) Time intelligence by Ship Date

Most time-intel functions follow the active relationship. To do the same logic by Ship Date, wrap them with USERELATIONSHIP:

Last Year by Ship Date

DAX

Sales LY (Ship Date) :=

CALCULATE (

[Total Sales],

USERELATIONSHIP (Calendar[Date], Sales[ShipDate]),

```
DATEADD ( Calendar[Date], -1, YEAR )

)
```

Month-to-Date by Ship Date

```
DAX

Sales MTD (Ship Date) :=

CALCULATE (

[Total Sales],

USERELATIONSHIP ( Calendar[Date], Sales[ShipDate] ),

DATESMTD ( Calendar[Date] )

)
```

Key idea: keep Calendar on the axis/filter; USERELATIONSHIP injects the ship-date relationship for the calculation window.

When to use USERELATIONSHIP

- **Role-playing dimensions** (Order/Ship/Invoice dates, Contract Start/End dates, etc.)
- You need **separate measures** for each date role without creating multiple calendar tables.
- You want **explicit and readable** control of which relationship applies to a measure.

Common pitfalls & tips

- USERELATIONSHIP works only **inside** CALCULATE (or functions that invoke a calc context transition).
- It **does not** permanently change the model—only the **current measure evaluation**.
- If you need **dynamic date role switching** in visuals without multiple measures, use a **disconnected slicer + SWITCH** as shown.
- For time-intelligence patterns (e.g., YTD/LY), always apply USERELATIONSHIP in the **same CALCULATE** that wraps the time-intel function.

- If you truly need separate independent date axes (rare), consider **separate Calendar tables**—but most cases are cleaner with USERELATIONSHIP.
- Avoid using it to "fix" a poor model. Ensure:
    - The relationships are **single-direction** from Calendar to Sales
    - Calendar[Date] is a **contiguous** date range (no gaps)
    - Your inactive relationship columns (ShipDate) have the **same data type** and appropriate granularity

---

Quick test dataset (optional)

You can paste this into Power BI as Enter Data (or load from CSV) to try:

Calendar (cover a range):
Date from 2024-01-01 to 2024-01-31

Sales

OrderID	OrderDate	ShipDate	SalesAmount
1	2024-01-03	2024-01-05	100
2	2024-01-05	2024-01-10	250
3	2024-01-10	2024-01-12	300
4	2024-01-12	2024-01-20	150

Create the active relationship on OrderDate, inactive on ShipDate, and add the measures above.

TL;DR stands for "Too Long; Didn't Read."

- USERELATIONSHIP(Calendar[Date], Sales[ShipDate]) inside CALCULATE lets you compute the **same measure** by an **alternate relationship** (e.g., ship date) without making it active model-wide.
- Perfect for **role-playing** date scenarios and **side-by-side** analyses (Order vs. Ship).

REMOVEFILTERS()

REMOVEFILTERS() is a **DAX table function** that clears filters from *whatever columns or tables you pass into it*.
But importantly, it **does not wipe all filters blindly** — it only removes filters on the *specified* columns/tables, and only in the *current filter context*.

Here's the full breakdown:

✅ **What filters does REMOVEFILTERS() actually remove?**

1. Column filters

If you pass a column, REMOVEFILTERS(Column) removes **all filter context on that specific column**, including:

- Slicer selections
- Visual-level filters
- Page-level filters
- Report-level filters
- CALCULATE modifiers
- Implicit filters from relationships (via row context transition)

Example:

DAX

Sales All Products =

CALCULATE([Total Sales], REMOVEFILTERS(Products[ProductName]))

This removes **only** ProductName filtering but keeps filtering from other columns (like Region, Date, etc.).

2. Table filters

If you pass a table, it removes filters from **every column in that table**.

DAX

Sales All Products Table =

CALCULATE([Total Sales], REMOVEFILTERS(Products))

This clears filters from:

- Products[ProductName]
- Products[Category]
- Products[Subcategory]
- Any calculated columns in the table
- Any auto-exist filters from composite selections

…but it leaves other tables' filters intact (e.g., Dates, Geography).

3. All filters through relationships (if table is removed)

By removing all filters from a table, it also removes:

- Propagated filters
- Filters via many-to-one relationships
- Filters via bi-directional relationships

Example: removing Products filters also removes **any filtering that came from related dimension tables**, because they ultimately filter Products.

❗ **What REMOVEFILTERS() does *NOT* remove**

❌ **It does not remove filters from *unrelated* tables.**

If Dates is not related to Products, removing Products does not clear Dates filters.

❌ **It does not remove filters on measures unless those filters come from the target tables/columns.**

❌ **It does not change row-level security (RLS).**

RLS *always* stays enforced.

❌ **It does not remove filters applied by variables inside the measure, unless those variables depend on the removed column/table.**

🔍 **Comparison vs similar functions**

Function	Removes Which Filters?	Notes
REMOVEFILTERS(Column/Table)	Only specified columns/tables	Equivalent to ALL() with cleaner semantics
ALL(Column/Table)	Same as REMOVEFILTERS()	Also returns a table; sometimes behaves differently in ambiguous models
ALLEXCEPT(Table, Columns)	Removes all filters *except the listed ones*	Opposite intent
ALLSELECTED()	Removes external filters but keeps user context	Very different behavior
KEEPFILTERS()	Adds instead of removing filters	Opposite of REMOVEFILTERS

🧠 **Important subtlety:**

REMOVEFILTERS() removes both direct and propagated filters

Example:

- Visual filters Category → filters Products table
- Relationship filters Products → filters FactSales

Then:

DAX

CALCULATE([Total Sales], REMOVEFILTERS(Products))

Clears **all** of that, including the propagation into FactSales.

🏁 **Summary: What filters does REMOVEFILTERS affect?**

It removes:

- Filters applied directly on the given columns/tables
- Filters coming from slicers, visuals, page/report filter panes
- Filters created by CALCULATE transition
- Filters propagated from related tables
- Auto-exist filters

It does *not* remove:

- Filters on other tables/columns you didn't specify
- RLS
- Filters on unrelated tables
- Filters created by measures unless tied to the target column/table

Toggle Slicer

PBIX kit you can use to spin up the toggle slicer scenario (**Order vs. Ship vs. Invoice**) in under 3 minutes.

⬇️ Download the kit

- **ZIP (all assets)**:

tiny_pbix_toggle_kit.zip
(Contains CSVs, DAX measures, and a README with step-by-step instructions)

- If you prefer individual files:
  - Calendar.csv
  - Sales.csv
  - DateMode.csv (optional)
  - DAX_Measures.txt
  - README_Tiny_PBIX_Kit.md

Note: I can't directly attach a .pbix from here, but this kit is designed so you can create it in Power BI Desktop with just a few clicks.

What's inside & how to use it (quick guide)

1. **Load the data**
   - Power BI Desktop → **Get Data → Text/CSV** → import Calendar.csv, Sales.csv, and optionally DateMode.csv.
2. **Create relationships (Model view)**
   - **Active**: Calendar[Date] → Sales[OrderDate] (single-direction from Calendar → Sales)
   - **Inactive**:
     - Calendar[Date] → Sales[ShipDate]
     - Calendar[Date] → Sales[InvoiceDate]
3. **Disconnected slicer**
   - If you didn't import DateMode.csv, create a calculated table:

DAX

```
Date Mode = DATATABLE(
"DateMode", STRING,
{ {"Order Date"}, {"Ship Date"}, {"Invoice Date"} }
)
```

4. **Add the measures**
   Open

DAX_Measures.txt and paste them into a Measures table. The key ones:

DAX

```
Total Sales :=
SUM ( Sales[SalesAmount] )

Sales (Order Date) := [Total Sales]

Sales (Ship Date) :=
CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[ShipDate] ) )

Sales (Invoice Date) :=
CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[InvoiceDate] ) )

Sales (Toggle Date Context) :=
VAR mode = SELECTEDVALUE ( 'Date Mode'[DateMode], "Order Date" )
RETURN
SWITCH (
mode,
"Ship Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[ShipDate] ) ),
"Invoice Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[InvoiceDate] ) ),
[Total Sales] // Default: Order Date
)

// Time-intel variants that respect the toggle
Sales YTD (Toggle) :=
VAR mode = SELECTEDVALUE ( 'Date Mode'[DateMode], "Order Date" )
RETURN
```

SWITCH (

mode,

"Ship Date", CALCULATE ([Total Sales], USERELATIONSHIP ('Calendar'[Date], Sales[ShipDate]), DATESYTD ('Calendar'[Date])),

"Invoice Date", CALCULATE ([Total Sales], USERELATIONSHIP ('Calendar'[Date], Sales[InvoiceDate]), DATESYTD ('Calendar'[Date])),

CALCULATE ([Total Sales], DATESYTD ('Calendar'[Date]))

)

5. **Build the visuals**
    - Add a **Slicer** → Date Mode[DateMode] (Order/Ship/Invoice).
    - Add a **Line chart**:
        - X-axis: Calendar[Date]
        - Y-axis: Sales (Toggle Date Context)
    - (Optional) Add **Sales YTD (Toggle)** to compare YTD trends per mode.

Why this works

- USERELATIONSHIP **temporarily activates** the alternate date relationship (Ship or Invoice) only for the **measure** you're evaluating—your model's active relationship stays on **Order Date**.
- The **disconnected slicer** drives a SWITCH() in the measure so users can change the date role **without** duplicating visuals or creating multiple calendars.
- Time-intelligence measures (YTD/MTD/LY) piggyback on the same toggle by wrapping the appropriate USERELATIONSHIP in the **same CALCULATE** that applies the time window.

- Add more toggles (e.g., **Invoice Paid Date**, **Delivery Date**) and comparisons (e.g., **delta between Order vs. Ship**).

YTD, **MTD**, and **Last Year (LY)** versions that respect the same **Order / Ship / Invoice** toggle.

↕ YTD/MTD/LY toggles

- **ZIP (v2)**:

tiny_pbix_toggle_kit_v2.zip
Includes Calendar.csv, Sales.csv, DateMode.csv, DAX_Measures_v2.txt, and README_Tiny_PBIX_Kit_v2.md.

(If you want the original v1 as well: tiny_pbix_toggle_kit.zip)

What changed

- New measures in **DAX_Measures_v2.txt**:
    - Sales YTD (Toggle)
    - Sales MTD (Toggle)
    - Sales LY (Toggle)
      All three **follow the slicer** (Date Mode[DateMode]) and use USERELATIONSHIP to activate **Ship** or **Invoice** date as needed.

Copy/paste measures (for quick reference)

You can also grab them directly here:

DAX

```
// Year-to-Date

Sales YTD (Toggle) :=

VAR mode = SELECTEDVALUE ( 'Date Mode'[DateMode], "Order Date" )

RETURN

SWITCH (

mode,

"Ship Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[ShipDate] ), DATESYTD ( 'Calendar'[Date] ) ),

"Invoice Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[InvoiceDate] ), DATESYTD ( 'Calendar'[Date] ) ),

CALCULATE ( [Total Sales], DATESYTD ( 'Calendar'[Date] ) )

)


// Month-to-Date

Sales MTD (Toggle) :=

VAR mode = SELECTEDVALUE ( 'Date Mode'[DateMode], "Order Date" )

RETURN

SWITCH (

mode,

"Ship Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[ShipDate] ), DATESMTD ( 'Calendar'[Date] ) ),

"Invoice Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[InvoiceDate] ), DATESMTD ( 'Calendar'[Date] ) ),

CALCULATE ( [Total Sales], DATESMTD ( 'Calendar'[Date] ) )

)


// Last Year (same period)
```

```
Sales LY (Toggle) :=

VAR mode = SELECTEDVALUE ( 'Date Mode'[DateMode], "Order Date" )

RETURN

SWITCH (

mode,

"Ship Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[ShipDate] ), DATEADD ( 'Calendar'[Date], -1, YEAR ) ),

"Invoice Date", CALCULATE ( [Total Sales], USERELATIONSHIP ( 'Calendar'[Date], Sales[InvoiceDate] ), DATEADD ( 'Calendar'[Date], -1, YEAR ) ),

CALCULATE ( [Total Sales], DATEADD ( 'Calendar'[Date], -1, YEAR ) )

)
```

Quick steps (recap)

1. **Load** Calendar.csv, Sales.csv, DateMode.csv.
2. **Relationships**: Active = Calendar[Date] → Sales[OrderDate]; Inactive = Calendar[Date] → Sales[ShipDate] & Sales[InvoiceDate].
3. **Paste** the measures from

DAX_Measures_v2.txt.

4. Drop a **Slicer** for Date Mode[DateMode] and use Sales (Toggle Date Context) or the new YTD/MTD/LY measures in visuals.

Parameters

1. Use Parameters to Visualize Variables - Power BI | Microsoft Learn
    - Create a parameter
    - Create a measure by using a formula that includes the parameter.
    - Create a visual that changes as you use a slicer to adjust the parameter value.
2. Filter a report using query string parameters in the URL - Power BI | Microsoft Learn
    - When you open a report in Power BI service, each page of the report has its own unique URL.
    - To filter that report page, you could use the Filters pane on the report canvas.
    - Or you could add query string parameters to the URL to prefilter the report.

Summary of Parameter Types

Parameter Type	Purpose	How it works
Numeric Range	"What-if" analysis (adjusting a single numerical variable)	Creates a slicer that inputs a numerical value into a DAX measure formula.
Fields	Dynamically changing the dimensions or measures in a visual	Creates a slicer that swaps the entire field used in a visual's axis or value well.

Quick chooser: which "parameter" should I use?

- **Switch server/database, file path, or API URL?** → *Power Query parameter*
- **Filter at the source with a slicer?** → *Dynamic M query parameter*
- **Let users switch measures/axes?** → *Field parameter*
- **Numeric scenario analysis (slider)?** → *What-if parameter*
- **Partition/refresh control?** → *Incremental refresh parameters*
- **Pixel-perfect reports with prompt inputs?** → *Paginated report parameters*

Power BI parameters can be used in several ways beyond the basic "what-if" pattern (create parameter ➞ use in a measure ➞ slicer controls value ➞ visual updates). Below is a concise map of the parameter types and the *other* places you can use them, with practical examples and gotchas.

1) Power Query Parameters (M) — for data shaping, environments, and governance

Where they work

- **Data source switching** (Dev/Test/Prod connection strings, database names, schema names)
- **Credentials and paths** (server name, SQL database, file paths, API base URLs)
- **Reusable constants in M logic** (e.g., filters, limits, column selections)
- **Dataflows** (same support in the Service; can be edited after publish)
- **Deployment pipelines** (map parameter values per stage via rules)
- **Incremental refresh** (tie to RangeStart/RangeEnd datetime parameters)

Examples

- **Environment switcher**

Power Query

// Parameter: Environment = "Dev" | "Prod"

// Parameter: DevServer = "dev-sql"

// Parameter: ProdServer = "prod-sql"

Source = Sql.Database(

if Environment = "Prod" then ProdServer else DevServer,

"SalesDW"

)

- **API base URL**

Power Query

// Parameter: ApiBaseUrl = "https://api.contoso.com/v1"

Source = Json.Document(Web.Contents(ApiBaseUrl & "/orders?top=1000"))

Notes

- These parameters are **editable in the Service** (Dataset settings ➜ Parameters) without republishing.
- They're not for DAX; they live in M (Power Query) and affect data retrieval/transformations.

2) Dynamic M Query Parameters — push filters down to the source

Bind a slicer to a field and have it **drive a Power Query parameter at query time** (for supported DirectQuery sources like SQL Server, Snowflake, Synapse, etc.). This lets the filter be **folded** to the source, reducing data movement.

Use cases

- Switching between **large fact partitions** (e.g., Region = US/EU)
- **Server-side search** / Top N (avoid importing entire dataset)
- **Role-based** or **tenant-aware** queries (with caution)

Example concept

- Create a PQ parameter SelectedCountry.
- Enable **Dynamic M Query Parameters** and bind SelectedCountry to a model column.

- Use it in your native SQL:

Power Query

```
Value.NativeQuery(
Sql.Database(Server, Db),
"SELECT * FROM Sales WHERE Country = @country",
[Parameters = [country = SelectedCountry]]
)
```

Notes

- Works best with DirectQuery and foldable sources.
- Unlike What-if, the slicer drives data retrieval, not just visuals.

3) Field Parameters — switch fields/measures dynamically (no DAX calc groups needed)

What they are: A model table that lets users switch **dimensions/measures** via slicer. Super useful for building one visual that can pivot among multiple KPIs or categories.

Use cases

- **Dynamic axis or legend** (e.g., Month vs. Quarter vs. Year)
- **Dynamic measure selection** (e.g., Revenue vs. Margin vs. Units)
- **Dynamic detail drill** (compact UX instead of multiple report pages)

Example

- Create a Field Parameter with [Revenue], [Gross Margin %], [Units].
- Put it in a slicer and into your visual's **Y-axis**. The visual switches measure as the slicer changes.

Notes

- Field parameters can be used across visuals, conditional titles, tooltips, etc.
- They complement what-if parameters; they don't store scalar values—**they switch fields**.

4) What-If Parameters — beyond a single measure

You already know the basic pattern, but they can do more:

- **Drive conditional formatting** (colors, data bars, thresholds)
- **Dynamic titles/annotations** (refer to parameter value)
- **Scenario selections** (pair with SWITCH logic across multiple measures)
- **Sensitivity analysis** (multiple what-ifs combined)

Example

Power Query

Price Sensitivity = GENERATESERIES(0.8, 1.2, 0.05) // creates a table

Selected Factor = SELECTEDVALUE('Price Sensitivity'[Price Sensitivity], 1)

Adj Revenue = [Base Revenue] * [Selected Factor]

5) Incremental Refresh Parameters — governed refresh windows

- Special **datetime parameters** RangeStart and RangeEnd filter your fact table in Power Query.
- Power BI uses them to create partitions and refresh only recent data.

Example M filter

Power Query

FilteredRows = Table.SelectRows(Source, each [OrderDate] >= RangeStart and [OrderDate] < RangeEnd)

Notes

- These are parameters, but with a specific meaning for partitioning/refresh.

---

6) Paginated Reports (Report Builder) — classic parameters & integration

- **True report parameters**: prompt users, cascade parameters, pass between reports.
- From Power BI, you can **drill-through** to a paginated report and **pass filters as parameters**.
- Excellent for **pixel-perfect** exports and **large-detail** tables.

---

7) URL & Button-driven "parameters" for navigation

While not Power Query parameters:

- **URL filters** (querystring) can pre-filter a Power BI report.
- **Bookmarks** + **Buttons** can simulate **scenario selection** or parameter presets.
- Use with **Field Parameters** to create "view modes" (e.g., Summary vs. Detail).

---

8) Dataflows & Fabric (if applicable)

- **Dataflows Gen1/Gen2** support PQ parameters for reusable transformations.
- Parameters help centralize **source/tenant/env** variables across many datasets.

---

9) Governance & DevOps

- **Deployment pipelines**: define rules to **map parameter values per stage** (Dev → Test → Prod).
- **Service APIs / XMLA**: admins can read/update parameter values at scale (automation).

Common Pitfalls & Best Practices

- **Name carefully** (e.g., Env_Server, Api_BaseUrl, RangeStart/RangeEnd); keep a parameter dictionary in your project README.
- **Don't overuse what-if tables**; too many can bloat the model. Prefer **field parameters** for switching semantics.
- **Use dynamic M query parameters** for **large DirectQuery models** to reduce data movement.
- **Secure by design**: PQ parameters are not a substitute for **RLS**. Don't rely on them to hide data—enforce RLS at the model/source.
- **Service edits**: Remember that **Power Query parameters** are editable post-publish—great for ops, but coordinate changes.

Use Power Query Parameters for Server/DB/File/API Switching

Power Query parameters live in the **M** layer and allow you to dynamically change your **data source**, **file paths**, and **API endpoints** without touching the DAX model.

Here's how to do it cleanly.

---

1) Create the parameter

Example: environment name parameter

1. Home → **Manage Parameters** → **New Parameter**
2. Name: Environment
3. Type: Text
4. Allowed values: List
    - Dev
    - Test
    - Prod
5. Current value: Dev

Optional helper parameters (recommended)

You can also create parameters for each environment's server:

- DevServer = "dev-sql.company.com"
- TestServer = "test-sql.company.com"
- ProdServer = "prod-sql.company.com"

Same idea for:

- Database names
- File paths
- API base URLs

2) Use the parameter inside your M code

Example: switching SQL Server based on environment

Power Query

```
// Choose server based on the Environment parameter
ServerName =
if Environment = "Prod" then ProdServer
else if Environment = "Test" then TestServer
else DevServer,

Source = Sql.Database(ServerName, "SalesDW")
```

``

Example: switching file paths

Power Query

```
Path =
if Environment = "Prod" then "C:\Prod\Files\"
else if Environment = "Test" then "C:\Test\Files\"
else "C:\Dev\Files\",
Source = Csv.Document(File.Contents(Path & "Sales.csv"))
```

Example: switching API base URL

Power Query

```
BaseUrl =
if Environment = "Prod" then ProdApi
else DevApi,

Source = Json.Document(
Web.Contents(BaseUrl & "/orders?top=100")
)
```

This lets you move the entire solution across environments without editing queries.

---

3) Publish and edit parameters in the Power BI Service

After publishing:

In the Service:

Dataset → **Settings** → **Parameters**

From here you can choose:

- Dev
- Test
- Prod

…then refresh the dataset.

This allows **environment switching without republishing** the PBIX file.

---

4) (Optional) Use deployment pipelines for automated mapping

In a Power BI Deployment Pipeline:

- Go to **Parameter Rules**
- Map:
    - Dev stage → Dev parameters
    - Test stage → Test parameters
    - Prod stage → Prod parameters

This gives you **automatic environment switching** during deployment.

---

5) Best Practices

✓ Use descriptive parameter naming

Environment, ProdServer, DevApiBaseUrl, etc.

✓ Do NOT hardcode the server/path in queries

Always reference the parameter.

✓ Use parameters for everything that might change

- SQL server
- Database
- API URL
- File folder
- Filename
- Authentication method

✓ Document all parameter values in a README or wiki

✓ Avoid using PQ parameters for row-level security

They aren't secure for hiding user-specific data.

Use report readers to change visuals

[Use report readers to change visuals - Power BI | Microsoft Learn]

Field parameters in Power BI enable users to dynamically change measures or dimensions (e.g., swapping sales for profit, or city for category) within a single visual using slicers. To enable this, activate the feature in **File > Options and settings > Options > Preview features**, then create a parameter via **Modeling > New parameter > Fields**.

Example, the report reader can dynamically update the bar chart and table by selecting a field in the horizontal slicers above the visuals

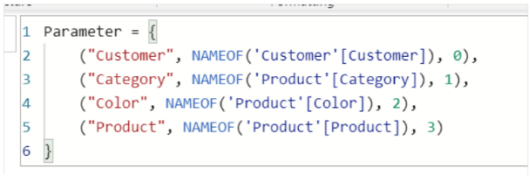
Once you create a field parameter, you can use the parameter to control the measures or dimensions used in a visual.

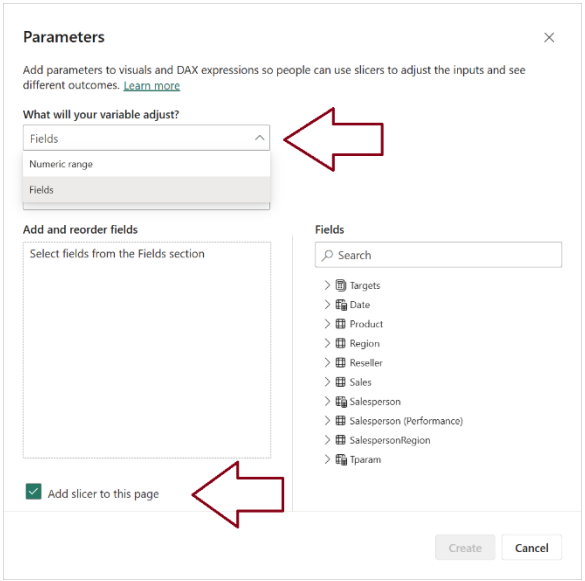
To edit an existing field parameter, modify the Data Analysis Expressions (DAX) directly.

For example, if you want to add a new field to an existing parameter, select **Shift + Enter** to start a new entry. Add a comma between entries, and match the format shown in this example. The number after the comma indicates the order of the fields:

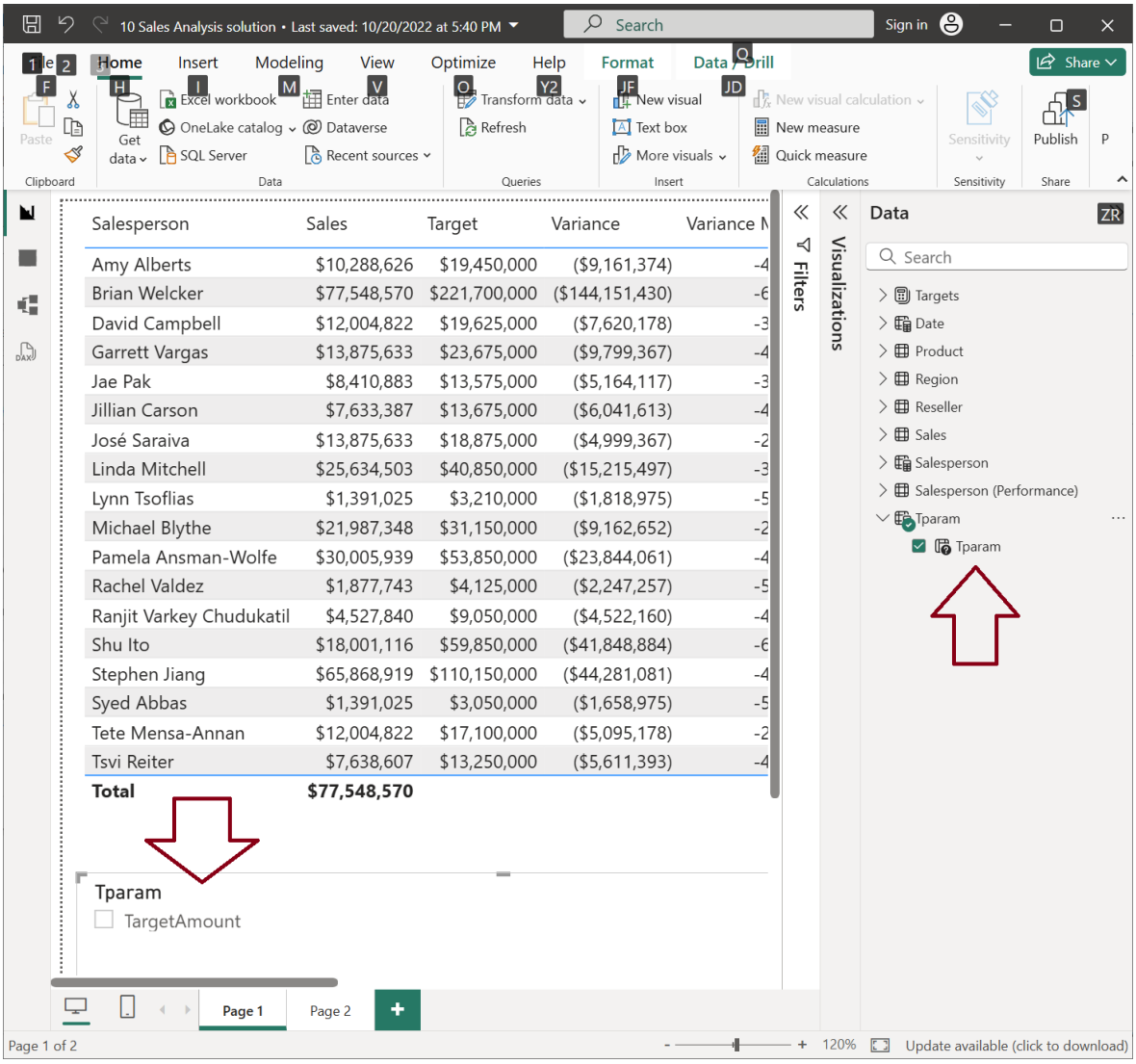
DAX Copy

```
Parameter = {
    ("Customer", NAMEOF('Customer'[Customer]), 0),
    ("Category", NAMEOF('Product'[Category]), 1),
    ("Color", NAMEOF('Product'[Color]), 2),
    ("Product", NAMEOF('Product'[Product]), 3)
}
```





When you create a parameter, a measure is also automatically created.



Steps to Create and Use Field Parameters:

- **Enable Feature:** Go to File > Options and settings > Options > Preview features and check **Field parameters**.
- **Create Parameter:** In the **Modeling** tab, click **New parameter > Fields**.
- **Add Fields/Measures:** Drag the desired columns or measures into the dialog box, name the parameter, and select "Add slicer to this page".
- **Apply to Visual:** Drag the new parameter field from the Data pane onto the Axis, Legend, or Values bucket of a visual.
- **Dynamic Interaction:** Use the generated slicer to select which fields appear in the visual.

Key Benefits:

- **Dynamic Visuals:** Allows users to change the axis, legend, or values (measures) of a visual dynamically.
- **Reduced Complexity:** Eliminates the need for complex DAX (SWITCH TRUE functions) or numerous bookmarks to create interactive reports.
- **Space Optimization:** Conserves report canvas space by reducing the number of visuals needed to display different data perspectives.
- **Mixed Data Types:** You can mix and match both measures and dimensions within the same field parameter.

Solved: Sort Visual on Field Parameter - Microsoft Fabric Community

allow a visual to switch between Date, Week and Month with Week and Month as Calculated Columns in a table.

allow a visual to switch between Date, Week and Month with Week and Month as Calculated Columns in a table.

Steps to Drill-Through from Power BI to Paginated Report

1. **Prepare Target Paginated Report (RDL):**
    - Open the detail report in Power BI Report Builder.
    - In the **Report Data** pane, create a parameter (e.g., ProductCategory).
    - Ensure the dataset query filters data using this parameter (e.g., WHERE Category = @ProductCategory).
    - Publish the report to a workspace in the Power BI service.
2. **Configure Action in Source Report:**
    - In the source Power BI Desktop report, select the visual (table, chart).
    - Right-click the text box or data point, select **Text Box Properties** or **Series Properties** -> **Action**.
    - Choose **Go to report** and specify the path to the published paginated report.
    - Click **Add** to map parameters:
        - **Name:** The exact parameter name in the target report (e.g., ProductCategory).
        - **Value:** The field from the current report to pass (e.g., [ProductCategory]).
3. **Finalize and Test:**
    - Publish the source report to the same workspace.
    - Clicking the linked data in the main report will now pass the filter and open the detailed paginated report.

Key Considerations

- **Parameter Matching:** Names must match exactly, including case.
- **Hidden Parameters:** You can set parameters to "Hidden" or "Internal" in the target report if you do not want users to change them after clicking.
- **Data Types:** Ensure the data type of the source field matches the target parameter type

Drillthrough Reports in a Power BI Paginated Report - Power BI | Microsoft Learn

Add a Drillthrough Action on a Power BI Paginated Report - Power BI | Microsoft Learn

Drillthrough Reports in a Power BI Paginated Report - Power BI | Microsoft Learn

Power BI URL and button-driven parameters

Use report readers to change visuals - Power BI | Microsoft Learn

example, the report reader can dynamically update the bar chart and table by selecting a field in the horizontal slicers above the visuals.

Power BI Embedded URL Generation: A Step-by-Step G... - Microsoft Fabric Community

generating secure embedded URLs for Power BI reports.

Power BI URL and button-driven parameters enable dynamic navigation and filtering by passing parameters via report URLs or using buttons with bookmark-driven actions. Key techniques include using the &pageName= URL parameter for deep linking, configuring buttons with actions, and utilizing Field Parameters to dynamically switch data.

1. Button-Driven Parameter Navigation (Bookmarks)

- **Create Bookmarks:** Set up visuals/filters on a page, then go to **View > Bookmarks** to add a new bookmark.
- **Add Button:** Go to **Insert > Buttons > Navigator > Bookmark navigator** or insert a blank button.
- **Assign Action:** Select the button, enable the **Action** option, set Type to "Bookmark," and select your created bookmark.

2. URL-Driven Parameter Navigation (&pageName=)

- **Find URL:** Open your report in the Power BI Service.
- **Get Page Name:** Copy the URL; the page identifier is after ReportSection.
- **Construct URL:** Append &pageName=ReportSection[Name] to the report URL to direct users to specific pages.

3. Field Parameters for Dynamic Navigation

- **Create Parameter:** Go to **Modeling > New parameter > Fields**.
- **Add Fields:** Drag fields to be swapped into the parameter dialog.
- **Use in Visuals:** Use the new parameter field in charts/tables. A slicer will automatically appear, allowing users to select the dimension.

4. Advanced: Passing Parameters to Other Reports

- **DirectQuery:** Use measures to concatenate a target report's URL with filtering parameters.
- **URL Encoding:** Ensure spaces and special characters in parameters are replaced with encoding (e.g., %20 for a space).
- **Suppress Elements:** Use &navContentPaneEnabled=false to create a cleaner, dashboard-like experience

Using dynamic M query parameters

[Dynamic M query parameters in Power BI Desktop - Power BI | Microsoft Learn]

Dynamic M query parameters give model authors more control over the filter selections to incorporate into DirectQuery source queries.

Using dynamic M query parameters with large DirectQuery models in Power BI can significantly enhance performance and control over data retrieval by incorporating filter or slicer selections directly into the source queries. This ensures only relevant data is retrieved, which is especially useful for very large datasets where loading the entire data is inefficient.

[DirectQuery in Power BI: When to Use, Limitations, Alternatives - Power BI | Microsoft Learn]

DirectQuery in Power BI lets you keep data in the source and query it at report time instead of importing it.

Quick decision guide

If you need	Consider first	Why
Maximum interactivity and full transformation flexibility	Import	In-memory columnar engine and rich modeling features
Near real-time changes on recent fact data plus historical context	Hybrid table (Import and DirectQuery partition)	Queries source for hot data and caches historical data.
Large lakehouse or warehouse scale with low latency reads (Fabric)	Direct Lake	Bypasses scheduled refresh and retains Import behaviors
Federated access to multiple external sources without full ingestion	DirectQuery (composite model)	Leaves data in place and blends sources.
Central governed enterprise model already published	Live connection to semantic model or Analysis Services	Reuses curated model and avoids duplication.
Push parameters to source at runtime (user-driven filtering)	DirectQuery with dynamic M parameters	Reduces scanned data and improves performance.
High concurrency and remote latency challenges	Import or Aggregations over DirectQuery	Aggregations accelerate common queries

DirectQuery Limitations w Dax

Solved: Re: List of DAX functions that do not work in Dire... - Microsoft Fabric Community

DAX Quick Measure Limitations with DirectQuery Tables

Certain **DAX quick measure calculations** are not supported when working with **DirectQuery tables** in Power BI due to performance and technical constraints. These limitations arise because some DAX functions used in quick measures translate poorly into SQL queries, which are executed directly on the underlying data source in DirectQuery mode.

Unsupported Quick Measures in DirectQuery Mode

1. **Time Intelligence Calculations**: Quick measures like *Year-to-date total*, *Quarter-to-date total*, *Month-to-date total*, *Year-over-year change*, and other time-based calculations are not supported. These rely on DAX functions such as TOTALYTD, TOTALQTD, and SAMEPERIODLASTYEAR, which cannot be efficiently translated into SQL queries.
2. **Calculated Columns and Tables**: DAX functions like FILTER and ALLSELECTED used in calculated columns or row-level security (RLS) rules are not supported in DirectQuery mode. This limitation applies because these operations require row-by-row evaluation, which is incompatible with DirectQuery.
3. **Complex Aggregations**: Measures involving advanced aggregations, such as those using SUMX, MAXX, or MINX with nested filters, may fail or perform poorly in DirectQuery mode. These calculations often require intermediate results that cannot be directly computed in SQL.

Workarounds and Considerations

- **Switch to Import Mode**: If possible, switch the data source to Import mode, which allows full DAX functionality without these limitations.
- **Simplify Calculations**: Redesign measures to use simpler DAX expressions that are compatible with DirectQuery.
- **Custom SQL Views**: Pre-aggregate or pre-calculate data in the source database using SQL views, reducing the need for complex DAX calculations.
- **Use External Date Tables**: For time intelligence, ensure the date table is marked as a "Date Table" in Power BI and use pre-aggregated time-based metrics.

Understanding these limitations is crucial when designing reports with DirectQuery to ensure compatibility and optimal performance.

known to be restricted or **partially supported** in DirectQuery (and often fail silently or return wrong results):

❌ **DAX functions/features with known limitations in DirectQuery:**

Category	Examples	Notes
Iterator functions	SUMX, AVERAGEX, FILTER inside calculated columns	Allowed in measures, but usage is restricted or sub-optimal in calculated columns or calculated tables
Time intelligence	SAMEPERIODLASTYEAR, PARALLELPERIOD, DATESYTD, DATESMTD, DATEADD, etc.	**Only work** if a proper Date table is marked and **fully imported**.
Calculated tables	Any ADDCOLUMNS, GENERATE, CROSSJOIN, etc.	Not supported when dependent on DirectQuery tables
Relationship functions	USERELATIONSHIP, TREATAS	Work in many cases, but sometimes break with composite models
Ranking functions	RANKX	Can fail or give poor performance in DirectQuery-only context
Advanced math / stats	MEDIANX, PERCENTILEX.INC, GEOMEANX, etc.	Often unsupported or returns errors
Custom aggregations	VAR, complex nested IF/SWITCH logic	Might get rewritten poorly or result in "Can't display visual"
IN operator	'Table'[Col] IN { "A", "B" }	This is **not** supported in DirectQuery to most sources (especially SQL Server)
Virtual tables in measures	SELECTCOLUMNS, UNION, EXCEPT	Can result in query folding failures or unsupported queries

The problem is that even when DAX "works", the generated SQL may be too complex or invalid — leading to errors or blank visuals.

2. **If one table is in DirectQuery mode, does that affect the whole model's DAX support?**

Yes — to a significant extent.

In **composite models (Import + DirectQuery)**:

- If your **measure or visual uses even one column from a DirectQuery table**, the **entire DAX query runs in DirectQuery mode**.
- This limits what DAX functions can be used — **even if most of your tables are Import**.
- Also, some features like **Auto Date/Time**, or certain calculated columns **cannot reference DirectQuery tables at all**.

Think of DirectQuery tables as "contaminating" any expression they touch — Power BI can't mix-and-match processing engines easily.

Recommendations

- **Offload logic to the source** if possible: Use SQL views or pre-aggregated tables for expensive DAX.
- **Import lookup/dimension tables** when using DirectQuery for fact tables.
- Use **Hybrid Tables** if using Premium or Fabric for recent + historical data split.

Step-by-Step Implementation

1. **Create Parameters in Power Query Editor**
    - In Power BI Desktop, go to **Home** > **Transform data** > **Transform data** to open the Power Query Editor.
    - Select **New Parameters** under **Manage Parameters** in the ribbon.
    - In the **Manage Parameters** window, create a new parameter (e.g., StartDateParam) with a default **Current Value**. Ensure the **Type** matches the data type of the column you will filter in your data source (e.g., Date/Time or Text).
    - Repeat to add other parameters as needed (e.g., EndDateParam).
2. **Reference Parameters in the M Query**
    - Select the DirectQuery table's query in the Power Query Editor.
    - Open the **Advanced Editor** from the **Home** tab.
    - Modify the M query to incorporate the parameters into the source query's filter clause. For SQL sources, this involves editing the native query (which may require turning off the security setting in Power BI Desktop Options to "Require user approval for new native database queries").
    - Example for T-SQL:
    - Sql.Database("YourServer", "YourDatabase", [Query="SELECT * FROM YourTable WHERE StartDate >= '" & Date.ToText(StartDateParam, "yyyy-MM-dd") & "' AND EndDate <= '" & Date.ToText(EndDateParam, "yyyy-MM-dd") & "'"])
    - Select **Done** to apply the changes.

    Create Tables of Values for Slicers

    - You need a separate, small, **Import** mode table for each parameter that contains the possible filter values for your report slicers (e.g., a table with various dates for StartDateParam).
    - In Power BI Desktop's **Modeling** tab, select **New table** and use a DAX expression to create this table (e.g., Date Slicer Table = CALENDAR(MIN('Table'[StartDate]), MAX('Table'[EndDate]))).

- Use a column name that does not exist in your actual data table to avoid ambiguity.

Bind Fields to Parameters

- Switch to the **Model** view in Power BI Desktop.
- Select the column in your new slicer table (e.g., the [Date] column in Date Slicer Table).
- In the **Properties** pane, expand the **Advanced** section.
- Under **Bind to parameter**, select the corresponding M query parameter from the dropdown list (e.g., StartDateParam).
- Configure **Multi-select** as needed. If set to No, the slicer must be in single-select mode.

Add Slicers to the Report

- Switch to the **Report** view and add a slicer visual using the field from your small slicer table.
- As report viewers interact with the slicer, their selections will dynamically update the M parameter, which in turn filters the DirectQuery source query at the source, improving performance.

Considerations and Limitations

- **Data Types:** Ensure the data type of the parameter and the bound column match.
- **Security Prompts:** You may need to disable the security prompt for native database queries during development.
- **Unsupported Features:** Dynamic M query parameters do not support aggregations, row-level security (RLS), or certain filter types like "Relative time slicer".
- **Publishing:** This feature is not supported on Power BI Report Server, but it works in the Power BI Service

Create and use parameters to visualize variables

[Use Parameters to Visualize Variables - Power BI | Microsoft Learn]

Create variables for your Power BI Desktop reports. Variables provide a way for you to visualize and quantify key values and then interact with those values in your reports. To adjust the value of a variable, you can use a slicer.

In Power BI, you can use **parameters** to visualize variables in two primary ways: **Numeric Range parameters** (for "what-if" analysis) and **Field parameters** (for dynamically changing visual dimensions/measures).

1. Using Numeric Range Parameters (What-If Analysis)

This method allows users to interact with a slicer to adjust a variable's value and observe the impact on a measure calculation.

- **Create the Parameter:**
    1. Go to the **Modeling** tab in Power BI Desktop.
    2. Select **New parameter** > **Numeric range**.
    3. In the dialog box, set the Name (e.g., "Discount Percentage"), Data type (e.g., Decimal number), Minimum, Maximum, and Increment values.
    4. Ensure **Add slicer to this page** is checked, then select **Create**. Power BI automatically creates a measure and a slicer.
- **Use the Parameter in a Measure:**
    1. Go to the **Modeling** tab and select **New measure**.
    2. Create a DAX formula that incorporates the new parameter's value.
    - *Example:* Sales after Discount = SUM(Sales[SalesAmount]) * (1 - 'Discount Percentage'[Discount Percentage Value])

4. Add the new measure to your visual (e.g., a column chart).

5. Interacting with the "Discount Percentage" slicer will now dynamically update the Sales after Discount value in the visual.

Using Field Parameters (Dynamic Visuals)

- This method lets report consumers dynamically change the fields (measures or dimensions) used in a visual's axes.
- **Enable the Feature (if necessary):** This is generally enabled by default, but if you don't see the option, go to `File` > `Options and Settings` > `Options` > `Preview Features` and check the box for **Field parameters**.
- **Create the Parameter:**

0. Go to the **Modeling** tab.
1. Select **New parameter** > **Fields**.
2. In the `Parameters` dialog, provide a `Name` (e.g., "Dynamic Axis").
3. Drag and drop the desired fields from the **Fields** pane into the `Add and reorder fields` section.
4. Check **Add slicer to this page** and select **Create**.

- **Use the Parameter in a Visual:**

0. Add a visual (e.g., a bar chart) to your report page.
1. Instead of a static data field, drag the new "Dynamic Axis" parameter from the `Fields` pane into the appropriate well (e.g., the `Y-axis` or `X-axis`).
2. The report user can now use the "Dynamic Axis" slicer to instantly swap between the available fields in the visual.

Step-by-step process Change Page Navigation & Button Text

✔ **Goal:**
If **[Error Rate] > 5%**, button shows **"Error Rate"** and navigates to **Error Rate Page**.
Otherwise, button shows **"On-time Rate"** and navigates to **On-Time Rate Page**.

✅ Step 1 — Create a DAX measure that outputs the *page name*

This measure returns the target page **as text**, based on the error rate.

DAX

```
Navigation Target =
IF (
[Error Rate] > 0.05,
"Error Rate Page", -- must match the real page name EXACTLY
"On-Time Rate Page"
)
```

Show more lines

Also create a second measure for the **button text**:

DAX

```
Navigation Label =
IF (
[Error Rate] > 0.05,
"Error Rate",
"On-time Rate"
)
```

Show more lines

✅ Step 2 — Insert a Button

1. Go to **Insert → Buttons → Blank**
2. Select the button.

✅ Step 3 — Set Button Type to *Page Navigation*

With the button selected:

1. Open **Visualizations → Button → Action**
2. Turn **Action = On**
3. Set **Type = Page navigation**

Now you'll see a field called **Destination**.

✅ Step 4 — Conditionally format the Page Navigation Destination

This is the key part.

1. Hover over **Destination**
2. Click the **fx** icon to apply conditional formatting
3. Choose:
    - **Format style:** *Field value*
    - **Based on field: Navigation Target**

Now the page navigation changes automatically based on the DAX value.

✅ Step 5 — Conditionally format the Button Text

1. Expand **Button → Style → Text**
2. Turn **Text = On**
3. Hover over the text box and click the **fx** icon
4. Set conditional formatting:
    - **Format style:** *Field value*
    - **Based on field: Navigation Label**

Now the button's label changes depending on [Error Rate].

📣 Result

Your button will now:

- Show **Error Rate** and navigate to the **Error Rate Page** when [Error Rate] > 5%

- Show **On-time Rate** and navigate to the **On-Time Rate Page** otherwise

Power BI Forecasting

a **clean, practical breakdown** of the **actual math** behind Power BI's Forecasting (line charts) so you know exactly how it calculates **units + seasonality** for **weekly, monthly, and multi-year** forecasts.

This is **not** Microsoft's proprietary code, but the math is public and documented because Power BI uses **ETS (Exponential Triple Smoothing)**, also called **Holt-Winters** — specifically the **ETS(A, A, A)** additive model.

✅ 1. Power BI Forecasting = ETS (Holt-Winters Triple Exponential Smoothing)

Power BI uses the statistical algorithm:

Forecast = Level + Trend + Seasonality

Mathematically:

1. **Level (L_t)**

$$L_t = \alpha(Y_t - S_{t-m}) + (1-\alpha)(L_{t-1} + T_{t-1})$$

2. **Trend (T_t)**

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

3. **Seasonality (S_t)**

$$S_t = \gamma(Y_t - L_t) + (1-\gamma)S_{t-m}$$

4. **Forecast h periods ahead**

$$\hat{Y}_{t+h} = L_t + hT_t + S_{t-m+(h \bmod m)}$$

Where:

- **Y_t** = your historical units
- **m** = season length (Power BI infers this when you choose Seasonality = Auto)
- **α, β, γ** = smoothing coefficients (Power BI automatically optimizes them)

✅ 2. What "Units + Seasonality Math" Means in Practice

Let's make this less abstract.

🔷 When Power BI sees weekly patterns

For **weekly** data:

- **m = 7**
  (Power BI detects 7-day cycles)

Seasonality term rotates every 7 data points.

🔷 **When Power BI sees monthly patterns**

For **monthly** data:

- **m = 12**
  (12 months in a seasonal cycle)

🔷 **When your dataset is years long**

If you forecast **years**, the model still uses:

- **m = 12** (monthly)
- or **m = 52** (weekly)
- or **m = 365** (daily)

depending on the underlying grain.

Power BI automatically picks this based on your axis granularity.

✅ **3. The key math differences: Week vs. Month vs. Year**

The formulas don't change —
only the value of m (season length) changes.

📆 **If forecasting weekly**

m = 7

Meaning: pattern repeats every 7 points.

Example: sales spike every weekend.

Forecast logic:

- Level: current baseline
- Trend: slope over the last several weeks
- Seasonality: 7-point repeating pattern

If forecasting monthly

m = 12

Example: seasonality repeats each year (Jan, Feb, Mar…).

Forecast logic:

- Level: baseline units
- Trend: long-term direction
- Seasonality: 12 repeating monthly adjustments

If forecasting across multiple years

Power BI still uses **single season length** based on the lowest axis grain.

If the X-axis uses **years as display**, but your data is monthly:

- Power BI uses **monthly seasonality (12)**
- Then aggregates the results into the chart

So the forecast is *still* based on monthly patterns.

✅ 4. How Power BI decides "seasonality" automatically

Power BI uses an autocorrelation-driven selection:

1. It computes candidate seasonalities using:
    - ACF (autocorrelation function)
    - MAPE reduction heuristics
2. Tests multiple m values
3. Picks the one giving best likelihood & lowest error

You can override this manually.

✅ 5. Practical example

Say you have **units sold** daily:

Y_t = units each day

m = 7 (weekly seasonality)

Power BI computes:

- L_t → baseline daily units
- T_t → recent trend (increasing or decreasing)
- S_t → repeating weekly cycle (Mon–Sun adjustment)

Forecast for 14 days:

$$\hat{Y}_{t+h} = L_t + hT_t + S_{(h \bmod 7)}$$

If sales spike on Friday, the forecasted Fridays will also spike.

✅ 6. TL;DR — What math drives Power BI forecasting?

Concept	Meaning
Model	ETS Triple Exponential Smoothing (Holt-Winters Additive)
Seasonality (m)	Based on axis granularity: 7=weekly, 12=monthly, 365=daily
Forecast	= Level + Trend + Seasonal component
What you control	season length, confidence interval, forecast horizon
What Power BI auto-optimizes	α, β, γ (the smoothing multipliers)

Shared Semantic Model Datasets

Depending on your data and report, you can do an adhoc data refresh anytime or set a schedule. However, scheduled refreshes require gateways.

"Shared dataset" behavior is essentially a Power BI Service/Fabric feature.
But there are a few **workable offline/desktop-only patterns** depending on what you need.

What you can do without the Service

1) Model-only PBIX (no report pages)

- You can absolutely build a **dataset/model-only** PBIX in Desktop (just delete report pages).
- You can **share the PBIX file** itself with someone else. When they open it in their own Desktop, they can create visuals **inside that file**.
- **Limitation:** This is *not* a "shared dataset." Each person is editing their own **copy** (no central model; no thin reports that point to a single dataset).

Good for: handoff of a reusable model as a file, one author at a time.

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2) Connect one Desktop to another Desktop's open model (same machine)

- In Power BI Desktop, **Get data → Power BI datasets** shows a **"This PC"** section for models currently open in Desktop on **that machine**.
- You can open a second Desktop instance and create a **thin report** that live-connects to the model in the first instance.
- **Limitations:**
    - This works only **on the same PC** (not a network-wide server).
    - The first Desktop instance must remain **open**; it's not a durable, shareable endpoint.
    - Not practical for team collaboration across machines.

Good for: a local "thin report" dev pattern or quick self-checks.

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3) SSAS Tabular (on-prem) as the shared semantic model

- If you want a **true shared dataset experience off the cloud**, use **SQL Server Analysis Services (SSAS) Tabular** on a server you control.
- Power BI Desktop can **live connect** to SSAS Tabular just like it does to a Power BI Service dataset.
- Multiple report authors can build thin reports against the same central model.

- **Tradeoffs:** Requires SSAS setup/ops, role/security management, and deployment pipeline for the Tabular model (e.g., via Visual Studio/Tabular Editor).

Good for: on-prem "shared dataset" equivalent without Power BI Service.

https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-analysis-services-tabular-data

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What is unique to the Service/Fabric

- **Shared datasets** that multiple thin reports connect to
- **Centralized discovery** (lineage, endorsement, certified datasets)
- **Row-Level Security** enforcement at scale with auditability
- **XMLA endpoint** for enterprise modeling (Premium/Fabric)
- **Deployment pipelines**, governance, and workspace permissions
- **Scheduled refresh** and managed gateways (for on-prem sources)

These are the cloud-native capabilities designed for **team-wide reuse** and lifecycle management.

Summary Matrix

Goal	Works with just files?	How to do it	Caveats
Reuse a dataset by handing off a file	✅	Share a **model-only PBIX**	Not a shared dataset; recipients edit their own copy
Build a thin report against another Desktop model	⚠️ Local only	Open PBIX #1 (model), then **Connect to "This PC"** from PBIX #2	Same PC only; the first Desktop must stay open
True "shared dataset" off cloud	✅ (server needed)	**SSAS Tabular**; connect Power BI Desktop live	Requires SSAS infrastructure & deployment process
True "shared dataset" in cloud	✅ Best practice	Publish dataset to **Power BI Service/Fabric**; build thin reports	Requires Service; brings governance features

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Practical recommendations

- If you just need to **hand someone a model** to build reports:
  → Save a **model-only PBIX** (or a **PBIT template**) and share the file.
- If you need multiple authors to **reliably reuse one central model without** the Service:
  → Stand up **SSAS Tabular** and have everyone live-connect to it.
- If you're okay using the cloud and want governance, RLS, pipelines, discovery, etc.:
  → Publish the dataset to **Power BI Service/Fabric** and build thin reports against it (the canonical pattern)

Dashboards/Workspaces

- Pinned visuals are set with filter context at time of pin.
- If the underlying visual changes, you'll need to update the dashboard tile as well.
- For time-based filters, it's a better idea to use a relative date slicer (or, Q&A using a relative time-based question).

	Dashboard	Workspace
Purpose	View & monitor KPIs	Create, manage, and store PBI content
What it contains	Pinned visuals from reports	Reports, dashboards, datasets, dataflows
Who can access	End users (view only)	Creators, contributors, admins
How users interact	Consume, share, set alerts	Build & edit content, schedule refresh
Update cadence	Updates when underlying report updates	Updates when content is modified or refreshed

RLS Row Level Security – Report vs Fabric/Service

Row-level security (RLS) with Power BI can be used to restrict data access for given users.

*Perform post-publication task to map security principals to the **Salespeople** role.*

- Filters restrict data access at the row level, and you can define filters within roles. In the Power BI service, users with access to a workspace have access to semantic models in that workspace.
- RLS only restricts data access for users with Viewer permissions. It doesn't apply to Admins, Members, or Contributors.
- You can configure RLS for data models imported into Power BI with Power BI.
- You can also configure RLS on semantic models that are using DirectQuery, such as SQL Server.
- For Analysis Services or Azure Analysis Services live connections, you configure row-level security in the model, not in Power BI.

The security option doesn't show up for live connection semantic models.

1. Securing Power BI with Advanced RLS & OLS | Rysun
2. Row-level security (RLS) with Power BI - Microsoft Fabric | Microsoft Learn - Service
3. Manage and secure Power BI - Training | Microsoft Learn – Report
4. Row-level security (RLS) with Power BI - Microsoft Fabric | Microsoft Learn – Model
5. Static vs Dynamic Options: userprincipalname() function will compare the email address from the Employees table with the email that the user entered when signing in to Power BI service.
6. SSO: Azure SQL Database for their sales operations that resides in the same tenant as Power BI. The database enforces RLS to control access to rows in various database tables. You can create a DirectQuery model that connects to this database without roles and publish it to the Power BI service. When you set the data source credentials in the Power BI service, you enable SSO. When report consumers open Power BI reports, Power BI passes their identity to the data source. The data source then enforces RLS based on the identity of the report consumer.
7. Object-level security (OLS) can restrict access to specific tables and columns, and their metadata.
8. The table includes columns that store employee name, phone, email address, and salary. While general report consumers can see employee name and contact details, they must not be able to see salary values. Only senior Human Resources staff are permitted to see salary values. So, the data modeler used OLS to grant access to the salary column only to specific Human Resources staff.

Object Level Security – RLS Advanced

1. Get started with Tabular Model Definition Language (TMDL) | Microsoft Learn
2. To turn on this public preview feature, go to File > Options and settings > Options > Preview features and check the box next to TMDL View.
3. When a user opens a Power BI report that queries a restricted object (for them), the error message could be confusing and will result in a negative experience. To them, it looks like the report is broken. A better approach might be to create a separate set of models or reports for the different report consumer requirements.

Azure Functions

1. Functions in multiple languages such as **C#, Java, JavaScript, Python, PowerShell, and Java**. Functions are triggered by events like HTTP requests, database changes, message queues, timers, or file uploads, and can connect to other Azure services through **bindings** without extra boilerplate code.
2. Azure Functions overview | Microsoft Learn
3. Azure Functions Scenarios | Microsoft Learn

Azure Dataverse

4. Add a Microsoft Dataverse database - Power Platform | Microsoft Learn
5. Licensing overview for Microsoft Power Platform - Power Platform | Microsoft Learn
6. Power Platform Licensing Guide – pdf
7. Connect to Dataverse with model context protocol (MCP) - Power Apps | Microsoft Learn
    - Model Context Protocol (MCP) is an open protocol that enables seamless integration between large language model (LLM) applications and external data sources and tools.
8. Billing rates and management - Microsoft Copilot Studio | Microsoft Learn

Centralized Data Ecosystem

case-study style comparison for how a **government entity** (GCC / GCC High / DoD) with **Azure**, **Microsoft Fabric**, and **Microsoft 365/O365** could build a **central data store** to power enterprise-grade **Power BI semantic models**.

The answer uses evidence from Microsoft-published and industry sources to ensure accuracy.

Case Study: Building a Central Government Data Store for Power BI Using Azure, Fabric & O365

A U.S. government agency wants to consolidate data scattered across line-of-business apps, legacy SQL systems, SharePoint lists, and spreadsheets. They already own:

- Azure Government subscription
- Microsoft Fabric capacity
- Microsoft 365 GCC / GCC-High
- Power Platform (Dataverse available)

Their goal: **Create a secure, governed, scalable central data store to service Power BI semantic models and reports.**

The agency is evaluating **three main options**:

1. **Microsoft Dataverse** (Power Platform SaaS data service)
2. **Azure SQL Database** (PaaS relational engine)
3. **Lake-centric architecture via Microsoft Fabric OneLake** (SaaS-based unified analytics platform)

1. Option A — Microsoft Dataverse

Dataverse is **not just a database**—it is a full **application data platform**: security model, API layer, business rules, and ALM included "out-of-the-box". [community....atform.com]

Why a government entity might choose Dataverse

- Built-in **RBAC**, auditing, encryption at rest & in transit, and compliance certifications (including **FedRAMP High**, DoD IL4/IL2 for Power Apps US Gov). [learn.microsoft.com]
- Strong integration with **Power Apps**, **Power Automate**, and **M365** (Teams, SharePoint integration, model-driven apps). [community....atform.com]
- Ideal for transactional workloads with business processes.

Limitations (particularly for large analytic workloads)

- Storage is more expensive than raw Azure services or lake storage. Additional storage is billed when quotas exceed allocations (tenant-pooled). [airbyte.com]
- Not designed for **high-volume analytics** like IoT, logs, or multi-TB fact tables.
- Performance/scale constraints vs. Azure SQL for deep analytical querying (Microsoft's own internal evaluation emphasizes this tradeoff). [microsoft.com]

Costs / Billing Units

- **Dataverse Developer Plan — free** (but not for production). [airbyte.com]
- **Dataverse Premium — $20/user/month**, includes 250 MB database + 2GB file storage per user, pooled at tenant level. [airbyte.com]
- Additional storage incurs **extra charges** when exceeding entitlements. [community....atform.com]

Is it free at rest?

❌ No — Dataverse storage beyond the included entitlements is a **paid resource**.

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2. Option B — Azure SQL Database (PaaS)

Azure SQL provides a **scalable relational engine** suitable for structured data, complex joins, and enterprise BI modeling. Government entities can run SQL in **Azure Government** with FedRAMP authorization (via LSP agreements). [azure.microsoft.com]

Why a government entity chooses Azure SQL

- Mature relational store for **enterprise data modeling** and serving semantic models.
- Multiple deployment models: **Single DB**, **Elastic Pools**, **Managed Instance** (supports more SQL Server features).

- Full VNet isolation, firewall restrictions, and network security capabilities for regulated workloads (e.g., CJI, FedRAMP). [it2edu.com]
- Predictable, high-performance analytical querying.

Limitations

- Requires DBA/data engineering skills.
- No low-code features (business rules, forms, workflows) like Dataverse.
- Governance/lineage must be built or handled in Fabric/Purview.

Costs / Billing Units

Billing is based on compute + storage:

- vCore model: compute + storage priced separately; **serverless** supports per-second billing for intermittent workloads. Azure SQL Database pricing is fully transparent and based on vCores, IO, and storage capacity. [azure.microsoft.com]
- DTU model: bundles compute and IO; monthly cost depends on tier (Basic, Standard S1/S3, Premium P1/P4) with example costs ranging from **$4.90 to over $1,860 per DB per month** depending on tier and storage allocated. [azurelessons.com]

Is it free at rest?

❌ No — Azure SQL is **metered** for compute + storage (GB/month).

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3. Option C — Microsoft Fabric OneLake (SaaS Lakehouse for Analytics)

Fabric unifies **data lake, warehouse, engineering, real-time analytics, and Power BI** into a single SaaS platform with built-in governance, metadata, security, and OneLake storage. [learn.microsoft.com]

Why a government entity chooses Fabric / OneLake

- Centralized "**single lake**" architecture enabling multi-department data sharing through OneLake federated storage and shortcuts (lakehouse + warehouse). [msfabric.pl]
- Strong native governance built into the platform, plus optional Purview integration for agency-wide governance and auditing. [learn.microsoft.com]
- Eliminates ETL duplication between lake and warehouse—great for Power BI semantic models.
- Works well for **multi-TB** datasets, logs, IoT, real-time feeds.

Limitations

- Not a transactional database; not a replacement for OLTP (Dataverse/SQL still needed for app write workloads).
- Requires Fabric capacity (SKU) — cost is fixed monthly, not usage-based.

Costs / Billing Units

- Governed by **Fabric Capacity Units (CU)**.
- OneLake storage is charged as **Azure Data Lake Storage Gen2** (standard storage rates apply) as Fabric rests on ADLS Gen2 per documentation. [msfabric.pl]
- Fabric governance features included; additional Microsoft Purview may require an extra license depending on features used. [learn.microsoft.com]

Is it free at rest?
❌ No — OneLake (Fabric) uses **metered Azure Storage costs** (hot/cool tiers depending on configuration), and Fabric capacity is billed.

Comparison Summary for a Government Central Data Store

High-Level Table

Requirement	Dataverse	Azure SQL DB	Microsoft Fabric (OneLake + Warehouse)
Best for	Apps, transactions, workflow, M365-integrated business data	Relational modeling, master data, BI semantic store	Lakehouse, analytics at scale, multi-modal data
Max Scale	Moderate (GBs)	High (TBs)	Very High (multi-TB to PB)
Compliance	FedRAMP High, IL4/IL2 (Power Apps Gov) [learn.microsoft.com]	Azure Gov FedRAMP	Fabric uses Azure security + Purview integration; supports gov compliance needs [p3adaptive.com]
Governance	Strong RBAC + built-in logic	Requires Purview / Fabric	Strong built-in governance + Purview option
Power BI Integration	Strong for business apps; moderate for enterprise BI	Excellent for semantic models	Native (Fabric Warehouse/Semantic models)
Cost Model	Per-user + storage	Compute + storage	Capacity units + storage
Free at Rest?	❌ No	❌ No	❌ No

Recommended Architecture for the Government Agency

Tiered Model (Best Practice)

1. Operational Layer (Write-heavy)

Use **Dataverse** or **Azure SQL** (depending on the system):

- Dataverse for Power Apps-based case management, permitting, inspections, HR workflows.
- Azure SQL for high-volume transactional systems, legacy migrations, master data hubs.

2. Analytical Lakehouse Layer

Use **Microsoft Fabric OneLake** for:

- Raw → cleaned → curated layers
- Data warehouse workloads
- Real-time analytics
- All data transformations
- Power BI semantic models (native in Fabric)

Fabric's design enables government agencies to **unify all analytics workloads**, modernize legacy data flows, and meet compliance frameworks for public sector environments. [p3adaptive.com]

3. Governance Layer

- Use **Fabric governance + Purview** for metadata lineage, DLP, classification, policy enforcement across the entire estate.
- Manage Fabric through the admin portal for tenant, domain, and workspace governance rules. [learn.microsoft.com]

Final Recommendation

For a U.S. government entity with Azure + Fabric + O365:

Use a hybrid approach:

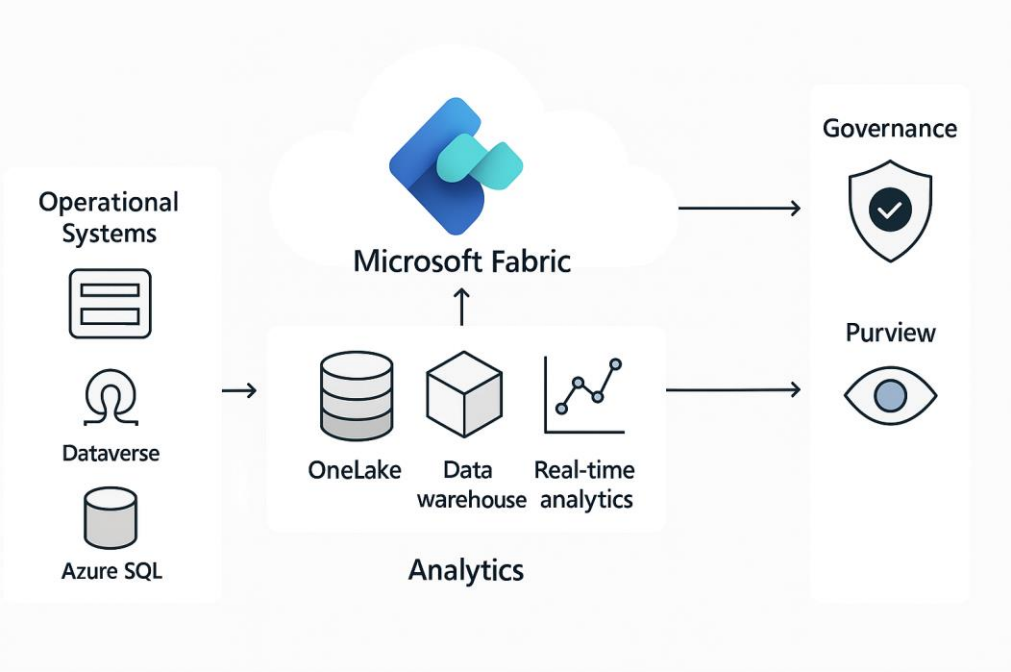
- **Dataverse** for low-code business processes, forms, auditing, secure RBAC, and workflow apps.
- **Azure SQL** for highly relational enterprise data and transactional workloads feeding the analytics layer.
- **Microsoft Fabric / OneLake** as the **central analytics store** powering **Power BI semantic models**, dashboards, real-time analytics, and data science needs.

This gives the agency:

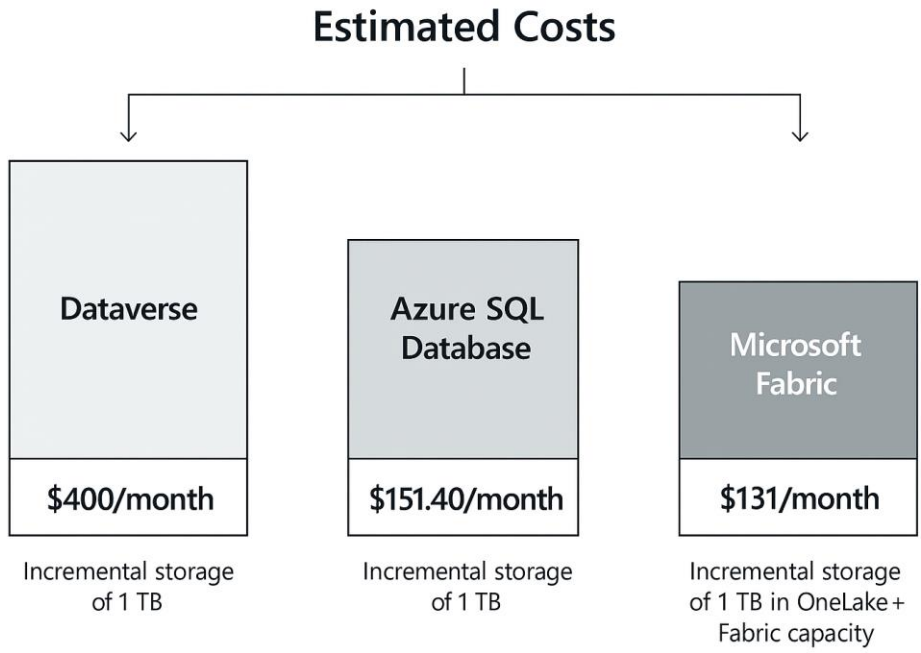
- Full compliance support
- Centralized, governed analytics

- Modernized lakehouse architecture
- Independence between transactional and analytics workloads
- Optimized costs

✅ An architecture diagram (Fabric + SQL + Dataverse integration)

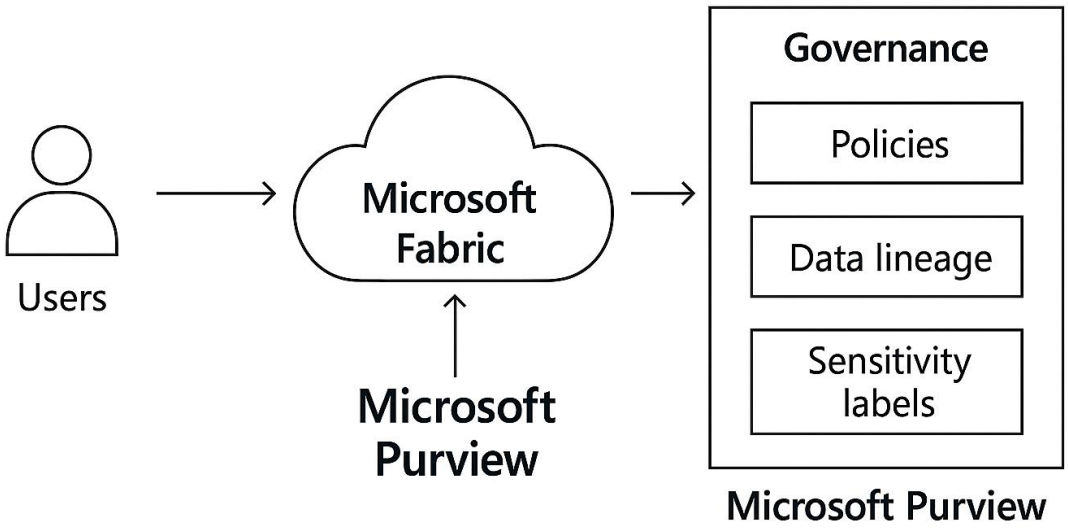


✅ A cost-estimation model comparing all three options



✅ A governance blueprint using Purview + Fabric

[Learn about data governance with Microsoft Purview | Microsoft Learn]



Reference

Exam PL-300: Microsoft Power BI Data Analyst - Certifications | Microsoft Learn

9. Course PL-300T00--A: Microsoft Power BI Data Analyst - Training | Microsoft Learn
10. GitHub - MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst
11. UseRelationship or Role-Playing Dimension; Dealing with Inactive Relationships in Power BI - RADACAD
12. Relationship functions (DAX) - DAX | Microsoft Learn
13. USERELATIONSHIP function (DAX) - DAX | Microsoft Learn
14. USERELATIONSHIP and direction of context propagation | Javier Guillén
15. Use Parameters to Visualize Variables - Power BI | Microsoft Learn
16. dotnet new <TEMPLATE> - .NET CLI | Microsoft Learn

More Filtering

17. USERELATIONSHIP and direction of context propagation | Javier Guillén

Templates

Azure Resource Manager templates are JavaScript Object Notation (JSON) files that define the infrastructure and configuration for your project.

18. AI app templates | Build AI applications with pre-made templates
19. ARM template documentation | Microsoft Learn

Analyzing & Large Report Example

1. **Data, Probability and Statistics (mathsisfun.com)
2. Ship INSIGHT (Container Ship) Registered in Marshall Is - Vessel details, Current position and Voyage information - IMO 9790907, MMSI 538007806, Call Sign V7ZZ9 | AIS Marine Traffic
3. **Data, Probability and Statistics (mathsisfun.com)
4. endlist